

ENRIQUE GABRIEL BAQUELA, PhD

OPERATIONS RESEARCH ENGINEER | OPTIMIZATION, SOFTWARE & AI FOR PLANNING

San Nicolás, Argentina | Remote | Occasional client travel | [LinkedIn](#) | [GitHub](#) | [Website](#) | [ORCID](#)

PROFESSIONAL SUMMARY

Industrial Engineer and PhD in Engineering with more than 10 years of hands-on experience designing optimization engines and decision-support software for logistics, port operations, manufacturing planning, scheduling, and resource allocation. Strong record of translating ambiguous business problems into mathematical models, algorithms, and software components used in operational environments. Deep expertise in LP/MILP, heuristics, metaheuristics, simulation, and ML-enhanced optimization. Primary algebraic modeling stack is Julia/JuMP, complemented by professional work in Python and C++, solver integration with HiGHS, SCIP and Gurobi, and working experience with PuLP and OR-Tools. Experienced in remote product teams, client-facing work, and end-to-end ownership.

CORE SKILLS

Optimization	LP/MILP; multi-objective optimization; assignment; routing; scheduling; network design; location; capacity and workforce planning
Algorithms	Exact methods; heuristics and metaheuristics; optimization via simulation; scenario and sensitivity analysis
Programming	Julia (advanced); Python; C++; SQL; R; LISP; modular package development; automated testing; Git/GitHub; Linux
Frameworks & solvers	JuMP; Gurobi; HiGHS; PuLP; OR-Tools; ConcurrentSim; AnyLogic; Vensim
AI for decisions	Supervised learning; deep learning; reinforcement learning; surrogate models; ML integrated with optimization and simulation
Industry domains	Supply chain; production planning and scheduling; maritime and port operations; multimodal logistics; inventory; energy; urban mobility

PROFESSIONAL EXPERIENCE

Lead Data Scientist & Operations Researcher — V2T Logistics

2020–2026

Singapore | Remote, multi-time-zone team

- Designed optimization and machine-learning engines for real-time logistics management solutions supporting operations for Heineken, CEVA, Shell, and Cargill in Asia.
- Developed LP/MILP and simulation components for routing, dispatching, capacity planning, multimodal networks, and finite-capacity manufacturing planning and scheduling.
- Integrated model inputs and outputs with SQL-backed product workflows and converted cost, service, and capacity trade-offs into scenarios and decision-ready visualizations.
- Owned work from business requirements and mathematical formulation through implementation, validation, stakeholder communication, and iterative improvement.

Technology: Julia/JuMP, Python, SQL, Gurobi, HiGHS, Linux

Founder & Operations Research / Data Science Consultant — Furgens Research S.A.

2014–Present

Argentina | Remote and client-facing

- Lead end-to-end consulting engagements for industrial companies, SMEs, and public-sector organizations in planning, scheduling, resource allocation, energy, logistics, and operational analytics.
- Built optimization and simulation-backed decision tools, including a steel-production scheduling prototype for Acindar-ArcelorMittal and an energy generation/consumption optimization service for Ternium Siderar.
- Manage problem discovery, data preparation, model design, software prototyping, validation, documentation, client training, and communication of business implications.
- Selected clients include Cargill, Ternium Siderar, Acindar-ArcelorMittal, ZTEC, and municipal governments.

Technology: Julia, Python, R, SQL, AnyLogic, Vensim, mathematical optimization and simulation

Data Scientist & Operations Research Specialist — Innovez One

2015–2020

Singapore | Remote

- Developed optimization engines for port operations, including pilot and tug assignment, resource scheduling, and vessel/barge planning for deployments in Singapore and Indonesia.
- Created university timetable optimization solutions using mathematical programming and heuristics.
- Worked with product and engineering teams to embed algorithms into production software and define operational metrics and visual decision support for planning users.

Technology: C++, LISP, Python, R, SQL, LP/MILP, heuristics, Linux

EARLIER PROFESSIONAL EXPERIENCE

Planning & Cost Control Engineer — Techint Engineering & Construction

2008–2014

Argentina | Atucha II and Ternium project sites

- Planned and controlled schedules, costs, and resources for large engineering and construction projects, coordinating with multidisciplinary teams, suppliers, and client representatives.
- Developed planning models, construction simulations, performance dashboards, and analyses of schedule and cost deviations to support corrective decisions.

Management Control Lead — Provser S.R.L.

2003–2008

San Nicolás, Argentina

- Managed budgeting, cost control, operational indicators, and process controls for an industrial-services SME, improving visibility and decision speed.

SELECTED TECHNICAL WORK

- FlowDES.jl (in development): modular Julia package and domain-specific language for discrete-event simulation, including entities, sources, queues, resources, routing, metrics, replications, and automated tests.
- Manufacturing planning: finite-capacity and sequence-dependent setup models with rolling-horizon optimization and operational data integration.
- Port logistics: resource-assignment and scheduling engines embedded in planning software for real operational environments.
- Optimization research: multi-objective math-heuristics, optimization via simulation, bilevel models, reinforcement learning, and surrogate-based optimization.

ACADEMIC LEADERSHIP — CONDENSED

- Full Professor of Operations Research, Universidad Tecnológica Nacional (UTN), 2023–Present; faculty member in Operations Research since 2006.
- Graduate and visiting faculty at Universidad de San Andrés, Universidad Austral, and UPAEP (Mexico), teaching operations research, optimization, simulation, machine learning, and supply-chain decision methods.
- Supervisor of doctoral and master's research in optimization, logistics, financial decision systems, workforce allocation, inventory, routing, and applied machine learning.

EDUCATION

Postdoctoral Research — Universidad Nacional de Rosario

2026–Present

- Research focus: integration of supervised learning and reinforcement learning within simulation-optimization methods.

PhD in Engineering — Universidad Nacional de Rosario

2020

- Dissertation: optimization-via-simulation techniques for traffic systems; thesis defense graded 10/10.

Industrial Engineer — Universidad Tecnológica Nacional (UTN)

2005

SELECTED PUBLICATIONS

- Baquela, E. G., & Olivera, A. C. "Hybridization of NSGA-II and MILP for optimization of the location of electric-scooter sharing stations." IEEE Latin America Transactions, 2022.
- Orlando, V. M., Baquela, E. G., Bhouri, N., & Lotito, P. A. "Public transport demand estimation by frequency adjustments." Transportation Research Interdisciplinary Perspectives, 2023.
- Baquela, E. G., & Olivera, A. C. "A novel hybrid multi-objective metamodel-based evolutionary optimization algorithm." Operations Research Perspectives, 2019.
- Full publication list available through ORCID and LinkedIn.

LANGUAGES & WORK STYLE

- Spanish: native. English: B2, Cambridge First Certificate in English (FCE).
- Remote collaboration since 2014; comfortable with asynchronous work, clear documentation, product/client communication, and occasional on-site visits.